

Assignment 2 and 3 Combined

WiDS

December 2025

1 Assignment 2

1. Consider a logistic regression model with a binary response variable

$$Y \in \{0, 1\},$$

and two predictors X_1 and X_2 . Suppose the dataset consists of two groups, with n_1 observations in group 1 and n_2 observations in group 2.

- (a) Write down the logistic regression model for Y in terms of the parameters β_0 , β_1 , and β_2 .
 - (b) Derive the likelihood function and the log-likelihood function for this two-group dataset.
2. Suppose we collect data for a group of students in a statistics class with the following variables:

X_1 = hours studied, X_2 = undergraduate GPA, Y = receives an A.

A logistic regression model is fit and the estimated coefficients are:

$$\hat{\beta}_0 = -6, \quad \hat{\beta}_1 = 0.05, \quad \hat{\beta}_2 = 1.$$

- (a) Estimate the probability that a student who studies for 40 hours and has an undergraduate GPA of 3.5 receives an A in the class.
 - (b) How many hours would the student in part (a) need to study in order to have a 50% chance of receiving an A in the class?
3. Suppose we wish to predict whether a given company will issue a dividend this year (Yes or No) based on

X = percentage profit in the previous year.

After examining a large number of companies, we observe the following:

- The mean value of X for companies that issued a dividend is $\bar{X} = 10$.
- The mean value of X for companies that did not issue a dividend is $\bar{X} = 0$.
- The variance of X for both groups is $\hat{\sigma}^2 = 36$.
- 80% of companies issued dividends.

Estimate the probability that a company will issue a dividend this year given that its percentage profit last year was $X = 4$.

4. Complete the Python notebook exercise on **Logistic Regression** provided on the Notion page. You are required to submit the completed code file.

Q.1

a.) as it is a logistic regression, we have a sigmoid curve :

$\therefore$

$$Y = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2}}$$

b.) couldn't accurately mathematically approach the functions.
Working on it:

Q.2

Given model:

$$\begin{aligned} a.) \quad x &= \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 \\ &= -6 + 0.05 X_1 + X_2 \end{aligned}$$

$$\text{for } X_1 = 40 \text{ \& } X_2 = 3.5 : \quad x = -6 + 0.05 \times 40 + 3.5 \\ = -0.5$$

$$\text{Probability : } P(Y) = \frac{e^x}{1 + e^x} = \frac{e^{-0.5}}{1 + e^{-0.5}}$$

$\therefore$ Prob. that student receives an A is 37.75%.

$$\text{Calc : } P(Y) \approx 37.75\%$$

b.) If $P(Y) = 50\%$ given $X_2 = 3.5$

$$\therefore 0.5 = \frac{e^x}{1 + e^x} \quad \therefore e^x = 1 \quad \therefore x = 0$$

$$\therefore -6 + 0.05 X_1 + 3.5 = 0$$

$$\therefore X_1 = \frac{2.5}{0.05}$$

$$X_1 = 50 \text{ hours}$$

Student has to study for 50 hours to achieve that probability.

Q.3 :

→ did not understand how to apply learnt probability theorems by blending with the given statistic information

Q.4: Link:

https://colab.research.google.com/drive/1Mk3_hDoucPBLVGfRjFxfjQqApBC6-8fN4?usp=sharing